THE UNBOUNDED DERIVED CATEGORY OF MODULES SUPPORTED IN A SPECIALIZATION-STABLE SUBSET IS THE FULL SUBCATEGORY OF COMPLEXES SUPPORTED THERE, IN FINITE DIMENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-local-cohomology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-local-cohomology/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

Editorial note. Bibliography provenance: neither the counted statement nor its proof contains a citation (this lane rescans the whole span before writing the card), so the card carries no bibliography and no entry was reconstructed.

Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The vanishing of local cohomology with support in a closed subset above the dimension of the ring. Author lemma lemma-cd-dimension, whose statement is reproduced verbatim as a context span of this card (source lines 758-764). Gives the bound on the cohomological dimension of the support functor.
- The colimit description of the sections with support in a specialization-stable subset, and the bounded below case of the equivalence. Author lemmas lemma-adjoint-ext and lemma-equal-plus, whose statements are reproduced verbatim as context spans of this card (source lines 996-1004 and 1019-1024). The first transfers the vanishing of the local cohomology to the general support, the second supplies the equivalence on bounded below complexes.

Lemma 0.1. *Let A be a Noetherian ring. Let $T \subset \mathrm{Spec}(A)$ be a subset stable under specialization. If $\dim(A) < \infty$, then functor $D(Mod_{A,T}) \to D_T(A)$ is an equivalence.*

Lemma 0.2. *Let $I \subset A$ be a finitely generated ideal of a ring A. If M is a finite A-module, then $H^i_{V(I)}(M) = 0$ for $i > \dim(Supp(M))$. In particular, we have $cd(A, I) \leq \dim(A)$.*

Lemma 0.3. *Let A be a Noetherian ring. Let $T \subset \mathrm{Spec}(A)$ be a subset stable under specialization. For any object K of $D(A)$ we have*

$$H^i(RH^0_T(K)) = \mathrm{colim}_{Z \subset T \text{ closed}} H^i_Z(K)$$

Date: stacks-project:local-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.4. *Let A be a Noetherian ring. Let $T \subset \mathrm{Spec}(A)$ be a subset stable under specialization. The functor $D^+(Mod_{A,T}) \to D^+_T(A)$ is an equivalence.*

Proof. Say $\dim(A) = d$. Then we see that $H^i_Z(M) = 0$ for $i > d$ for every closed subset Z of $\mathrm{Spec}(A)$, see Lemma 0.2. By Lemma 0.3 we find that H^0_T has bounded cohomological dimension.

Let $K \in D_T(A)$. We claim that $RH^0_T(K) \to K$ is an isomorphism. We know this is true when K is bounded below, see Lemma 0.4. However, since H^0_T has bounded cohomological dimension, we see that the ith cohomology of $RH^0_T(K)$ only depends on $\tau_{\geq -d+i}K$ and we conclude. Thus $D(\mathrm{Mod}_{A,T}) \to D_T(A)$ is an equivalence with quasi-inverse RH^0_T. □